Econ 414: Discussion Section 5

GSI: Brian Daza
bdaza@umich.edu

February 13th
University of Michigan

Today's Discussion

Review the answer to following questions through the lenses of the microfounded Romer model:

- Why does innovation generate profits?
- Why is monopoly power necessary for growth?
- Why do profits grow at rate g_L?
- How does free entry determine s_R?
- What determines equilibrium R&D?

I'm also providing you:

- a flowchart of the model to help us keeping ideas organized.
- a checklist on technical things you need to review.

Question 1: Why Monopoly?

Intermediate firms charge:

$$p = \frac{r}{\alpha}$$

Profits per firm:

$$\pi = (1 - \alpha)\alpha \frac{Y}{A}.$$

(a) What happens to the markup and profits as $\alpha \to 1$?

(b) What would happen to innovation under perfect competition?

(c) Why is monopoly power essential for endogenous growth?

Question 2: Growth of Profits

Profits per firm:

$$\pi_t = (1-\alpha)\alpha\frac{Y_t}{A_t}$$

On a balanced growth path:

$$g_Y = g_L + g_A.$$

(a) Show that $g_\pi = g_L$.

(b) Provide intuition: why do profits grow with population growth and not with technology growth?

Question 3: Free Entry and R&D

Free-entry condition:

$$\frac{s_R}{1-s_R} = \frac{\alpha(1-\alpha)}{1-\alpha}\frac{g_A}{r-g_L}$$

Without doing algebra, answer:

(a) What happens to R&D if r increases?

(b) What happens if g_L increases?

(c) What happens if monopoly power increases (i.e., α decreases)?

(d) What happens if patent protection becomes stronger?

Three Core Takeaways

1. No monopoly $\Rightarrow$ no profits $\Rightarrow$ no innovation.
2. Profits grow with market size (g_L), not with technology.
3. Free entry pins down how much labor goes into R&D.

Question 4: Labor Allocation and Tradeoffs

Recall:

$$L = L_Y + L_R, \qquad s_R = \frac{L_R}{L}.$$

(a) If more labor goes into R&D today, what happens to current output Y?

(b) Why is growth not "free" in this model?

(c) How does this differ from the Solow model?

Question 5: Market Size and Scale Effects

We showed that:

$$g_\pi = g_L.$$

(a) Why does a larger population make innovation more valuable?

(b) What does this model predict about large countries vs small countries?

(c) Is this prediction realistic? Why or why not?

Question 6: Policy and Growth

Suppose the government:

(a) Increases patent protection.

(b) Subsidizes R&D wages.

(c) Taxes monopoly profits.

For each case:

- What happens to V?
- What happens to s_R?
- What happens to g_A?

Question 7: Mechanism Mapping Exercise

Using the circular flow diagram provided, trace the chain:

$$\alpha \downarrow$$

Step-by-step:

1. What happens to markups?
2. What happens to profits π?
3. What happens to value V?
4. What happens to entry?
5. What happens to s_R?
6. What happens to growth?

Question 8: Where Do the Payments Go?

From the zero-profit condition of the final good firm:

$$Y = wL_Y + \sum_{j=1}^{A} p_j x_j$$

(a) Use $\pi_j = p_j x_j - r x_j$ to rewrite $\sum_{j=1}^{A} p_j x_j$ in terms of total profits and total capital.

(b) Using symmetry and $x_j = K_j$, show that:

$$Y = wL_Y + A\pi + rK.$$

(c) From lecture we obtained:

$$wL_Y = (1-\alpha)Y \quad \text{and} \quad A\pi = (1-\alpha)\alpha Y.$$

What does this imply for $\frac{rK}{Y}$?

(d) Why does wL_R not appear in this accounting identity?

Technical Checklist I: Micro Foundations

To manipulate the Romer model, you should be comfortable with:

- Deriving firm FOCs under monopolistic competition.
- Using elasticity to derive markups:

$$p + \frac{\partial p}{\partial x} x = r \quad \Rightarrow \quad p = \frac{r}{\alpha}$$

- Computing profits from revenues and costs:

$$\pi = (1 - \alpha)\alpha \frac{Y}{A}$$

- Understanding symmetry: $x_j = K/A$.
- Interpreting markup $\frac{1}{\alpha}$ and its economic meaning.

Technical Checklist II: Growth, and Equilibrium

To fully manipulate the Romer model, you should be comfortable with:

- Computing growth rates of ratios:

$$g_\pi = g_Y - g_A$$

- Using balanced growth logic:

$$g_Y = g_L + g_A \quad \Rightarrow \quad g_\pi = g_L$$

- Understanding the economic meaning of:

$$V = \frac{\pi}{r - g_\pi}$$

  - r = discount rate / opportunity cost
  - g_π = growth of profits (market size)

- Using free entry to determine s_R:

$$V = \text{Cost of innovation} \quad \Rightarrow \quad s_R \text{ determined in equilibrium.}$$

Technical Checklist III: Valuation

You should also be comfortable with:

- Continuous-time present discounted value:

$$V = \int_0^{\infty} \pi_0 e^{(g_\pi - r)t} dt = \frac{\pi_0}{r - g_\pi} \quad \text{(if } r > g_\pi\text{)}$$

If you can derive V and explain $r - g_\pi$, you understand the engine of growth in this model.

Next Week...

In the Romer model:

$$\text{Innovation} \Rightarrow \text{More Varieties}$$

But in reality:

- Smartphones replaced flip phones.
- Streaming replaced DVDs.
- AI may replace certain tasks.

What happens when innovation destroys existing firms?